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Review Article

Perceptuo-cognitive development in infancy and childhood

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Abstract

This paper reviews the development of perceptual and cognitive abilities during infancy and childhood as reported in the literature to connect them with the same in adults. First, infants are engaged with encoding perceptual information about objects and events in memory. When they acquire language skills, they begin to associate words with these perceptual representations and to encode narratives they have heard. This enables them to communicate to others what they have encoded and to reflect on it. Integrating perceptual encounters with emotional loading renders them personally relevant and enforces their encoding. However, the encounters experienced before the age of four years can rarely be recalled in adolescence or adulthood. Thus, beliefs formed during infancy remain largely inaccessible to memory recall later in life, although they are fundamental for individuals' behavior and the development of a person's perspective.

Keywords: belief; childhood; encoding; infancy; language; memory; perspective

1. Introduction

In this review we aim at elaborating the developmental milestones of perceptual, cognitive and neural information processing in relation to internal affective states in infants and children. This novel approach integrates working memory, social interaction, language capacities and long-term memory as an over-arching framework that affords shaping an individual's behavior. This framework is based on the psychomotor, perceptual and cognitive development in infancy and early childhood that has been the object of scientific study for many years, as reviewed recently [1]. When infants begin to interact with their environment, they associate their external perceptions with their internal states. For example, in the first weeks after birth infants learn to establish eye contact with their mother during feeding and learn that this connects them emotionally. They also experience that the tactile sensation of sucking their mother's nipples goes along with the cessation of hunger. Moreover, caregivers and others interacting with an infant typically smile and respond enthusiastically, especially when the infant reciprocates these behaviors. In such conditions, infants appear to develop the intuition that the people in their immediate environment are trustworthy and eager to satisfy their needs. At the age of a few months, infants start to sample sensory information about the property and quality of the objects in their environment—first those that are closer and later also those that are farther away [1]. These early representations of the environment are typically emotionally charged and, thereby, become relevant to the infants. All this information is encoded as neural representations in their brains.

As infants are known to possess only the capacity to execute mass movements of their limbs, including their hands, they obviously have difficulties in exploring and handling small objects. This changes by the age of six months when exploratory finger movements evolve, a process which continues beyond the age of 24 months [2, 3]. Moreover, as children grow physically, they become able to reach for objects above them and to observe events behind an obstacle—tasks that were insurmountable to them when they were small. Likewise, their hands get larger, with greater muscle strength, which allows them to routinely handle larger and heavier objects. Furthermore, the brains of children continue to grow and reorganize in a dynamic and non-linear fashion up to the age of nine years, as documented by diffusion imaging [4]. Due to this development, the networks in the brain become more integrated with increasing strength and efficacy and a decrease in functional modularity. These developmental changes are likely to affect the initial representations of infants and children concerning the properties of objects and events such that they will be updated in proportion to these physical changes.

During the various interactions with their caregivers, infants are also exposed to nursery rhymes and verbal phrases well before they become capable of speaking themselves. Later children hear narratives such as nursery rhymes, fairytales, and biographical stories. Although infants and children are typically touched by the emotional loading provided by such narratives, the proportion of verbal information that is understood can be expected to differ between infants below the age of 24 months and children beyond this age. Moreover, as verbal abilities continue to develop, children become increasingly able to articulate what they think and believe. In addition, they become aware of the feedback they receive. Furthermore, by exchanging views, children gradually become members of communities and broader social groups [5].

Nevertheless, it is important to emphasize that people typically do not have memories of their infancy and early childhood and can recall continuous memories only after the age of four years [6]. This phenomenon has been termed “infantile amnesia”. Notably, the few memories that go back to the age of approximately four years were reported to have a strong emotional content [7]. Accordingly, adults cannot recall most of their memories that were formed in infancy and early childhood, although they were the basis of their actions during childhood and probably also beyond. As many of the initial representations continue to be modified by new encounters during adolescence and adulthood, it is a puzzle to understand how they determine social and cultural behavior.

In this paper we address this puzzle by reviewing the evidence that is available in the literature. First, we describe the milestones of perceptual and cognitive development during infancy, i.e., up to 24 months, and during early childhood up to the age of four years. Then, the corresponding neural correlates are presented. This is followed by a section that relates perceptuo-cognitive and emotional development to belief formation in infants and children. Finally, we discuss with regard to the discourse with the humanities the development of the subjective perspective in infancy and childhood for the control of behavior.

2. Methodology

This theoretical paper aims at establishing a developmental framework for perceptuo-cognitive abilities including the formation of beliefs in infancy and childhood. Because there are large numbers of empirical studies on many specific aspects of the different capabilities pertinent to this goal, we capitalized on reviews and meta-analyses that were published—preferentially in the last ten years—and are listed in PubMed. The reason for this was that it would not be possible to cite every empirical paper on these topics within the limits of this paper, nor would any attempt for selection be balanced.

3. Perceptuo-cognitive development

Perceptual and cognitive abilities concern a number of different cerebral functions that will be addressed below. They have been reported to develop gradually in relation to the ontogenetic development of the body and, in particular, the brain. This development starts in the fetus as was described recently. For example, at about 25 weeks of gestation painful stimuli can be processed, and at the 34th gestational week the fetus was found to be able to discriminate between different sounds [8]. Specifically, music and speech can form stimulus-specific memory traces and influence neonatal autonomic and neural reactions to different sounds [9]. Within the first years after birth, children continue to develop a number of cognitive competencies that have been shown to already be quite mature [10].

3.1. Working memory

Probably the most basic cognitive competence is the engagement of working memory that enables infants to acquire the multimodal properties of objects and to assess events (**Figure 1**). This capacity has been shown to be independent from other cognitive abilities such as number approximation [11]. The representation of objects in working memory has been stated to encompass the number of items and the number of details that each of these items can hold. Both facets have been shown to increase in early childhood [12]. A meta-analysis of twelve reports has revealed that infants aged 10 to 20 months have an information processing limit of maximally three objects [13]. Children of the age of about 2.5 years were shown to have the ability to remember three objects in their locations if the objects were all unique, but failed when the objects shared features such as shape or color [14]. Notably, infants of only four months of age have a sensitivity to discriminating short temporal sequences but fail with sequences with long durations [15]. This points to a limited capacity for assessing temporal events. Moreover, 10- to 12-month-old infants could identify that four syllables were the same, while they failed to do so when five to six syllables were presented to them [16]. This suggests that infants of this age group can process short verbal expressions but encounter difficulties with narratives. Using a delayed spatial alternation task it was found that the working memory capacity improved with increasing age in infants [17]. This has been shown to continue during the elementary school period through to adolescence and adulthood [18].

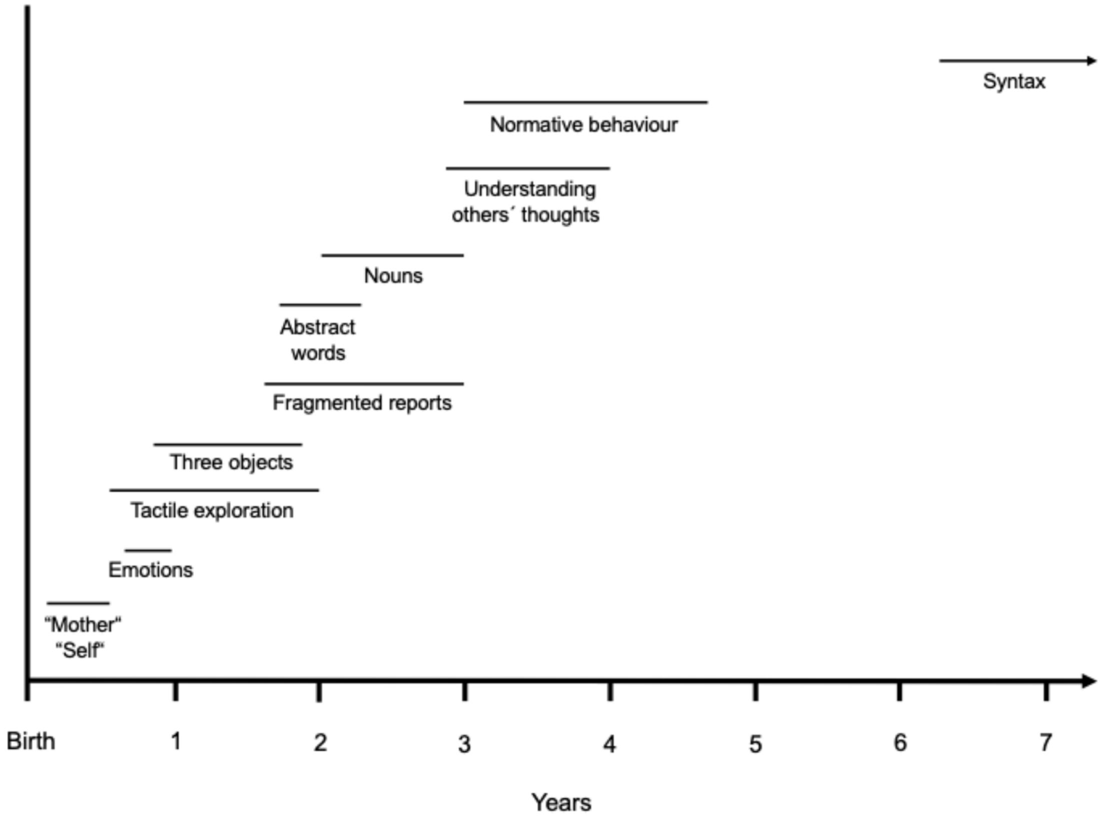

Figure 1. Timeline of the milestones of perceptuo-cognitive development in infants and children as evident from recent empirical studies: Recognition of the concept of "mother" and "self" [19] and categorization of emotions [20] occur within the first year. Tactile exploration [2, 3] and online processing of three objects [13] emerge in the second year. Learning of abstract words [21], fragmented reports about experiences [22] and saying all words known [23] occur after the age of two years. Understanding the thoughts of others [19] and knowing how to behave according to social norms [24] manifest between three and five years. Syntactic proficiency evolves at the age of seven to eight years [25].

3.2. Social interaction

The second competence to be acquired in infancy is social interaction. From birth onward, infants engage in innumerable social encounters and actively explore their environment long before they are able to express themselves verbally [1]. By as early as ten weeks of age infants respond distinctively to their mother's facial expression of joy; however, they do this in a more or less reciprocal fashion (**Figure 1**). But between only 2 and 6 months, infants develop the ability to act spontaneously, which is suggestive of a sense of self [19]. Moreover, between 3 and 4 years, children develop meta-cognitive abilities such that they understand that other individuals have thoughts and desires that may differ from their own [19]. This corresponds to a theory of mind ability being fully developed in children at the age of six years [26]. Interestingly, it is tightly related to the subjective experience of hope that has been reported to evolve in early childhood [27]. It was argued that hope results from the tension between existential fear and fundamental trust in caregivers. Importantly, early in life infants must establish trust in their primary caregivers, who reliably provide the care necessary for their survival [28]. For example, one of the first social achievements of an infant is their ability to let the mother or caregiver out of sight without fear and panic. Accordingly, the child apparently experiences consistency and continuity and is likely to develop trust in this person. Based on such early experiences children have been said to become socialized [29]. At the age of three to five years children also understand what they are expected to do and what to refrain from doing, as has been observed in a similar fashion among different cultural environments [24].

3.3. Language

A third cognitive competence is that children develop the capacity to verbally express what they think and feel (**Figure 1**). While events experienced in infancy before the age of about 18 months do not seem to be verbally accessible, events experienced in early childhood between 18 and 36 months can be reported in a fragmentary fashion [22]. During this same time, children realize that their desires are highly emotional for them [19]. However, only from around 36 months of age can children provide coherent accounts of their past experiences and retain these memories over extended periods [22]. This development was described as consisting of four components: (1) a multimodal input of sensory signals from the environment, (2) active adaptive learning by repetitive interaction with objects and people in the environment, (3) building of neural representations in the brain that facilitate recognition of the environmental input, and (4) adaptive learning during the dynamic developmental change [30]. In fact, there seems to be a bottom-up construction of neural representations in the brain as well as a top-down modulation of the interaction with the environment in the iterative process of knowledge acquisition. In serial reproduction experiments, words acquired earlier in life were found to be more concrete, more arousing, and more emotional and, therefore, more likely to survive retellings [31]. Moreover, in large retelling studies it was found that appraisals of happiness, sadness, and embarrassment showed high preservation, whereas disgust and risk were not preserved [32]. Thus, emotion appraisals seem to serve as anchors for the remembering and retelling of stories.

However, the acquisition and use of words by children have been found to be gradual. At about 7 to 8 months children begin to segment words from larger units of speech, which has been found to be predictive of their vocabulary at about two years [33]. That is, children individuate single words from continuous speech and associate them with intended referents in the environment [34]. By 12 months, children produce their first signs or words and by 24 months children link signs and words [29]. At the age of 18 to 24 months, children were shown to learn words best when their caregivers used gestural cues and when there were more potential referents and there was referential ambiguity for words [34]. At about two years, children say many fewer words than they understand and use no verbs, although they understand many. This is followed by a phase when children say all the nouns they understand and begin to use verbs, which coincides with the increasing capacity of the child's semantic memory [23].

A child's overall vocabulary is likely to reflect the range of concepts a child can hold. For example, at the age of 22 to 24 months there is a prominent progression in the acquisition of words denoting abstract concepts [21]. Words acquired between 2 and 4 years are more salient and easier to process than words acquired at the age of 7 to 8 years. In addition, the words acquired earlier are more stably represented within the community's language [35]. Generally speaking, concrete words are rooted in sensory experience, whereas abstract words are grounded in their associative use in language [36]. Moreover, abstract words with emotional valence have been found to be learned better than neutral abstract words by children up to the age of 8 to 9 years [37]. Word valence, interoception, and mouth action facilitate abstract word acquisition more than concrete word acquisition, showing that emotion and oral sensorimotor activity are important for abstract word acquisition in children [38]. The list of over 40.000 words the authors used in one study included nouns, adjectives and verbs. Importantly, more positive and more negative abstract verbs were found to be acquired at an earlier age than neutral abstract words [39]. Although sensorimotor experience is central to children's learning of concrete words, the combination of social and emotional experience has been shown to facilitate the acquisition of both concrete and abstract words [40]. By the age of 8 there is a significant increase in abstract words in the child's lexicon and proficiency in syntactic ability [25]. Conversely, children developing normally are more accurate in defining concrete and abstract words than children with a developmental language disorder [41]. Specifically, children were found to have a conceptual understanding of mercy in religious contexts but not in secular contexts [42]. One should be aware, however, that words are arbitrary and conventional sounds that signify objects and events, whereas their meanings are determined by social agreement [43].

3.4. Long-term memory

The fourth cognitive capacity is the encoding of information and the capability to actively recall it. Long-term memory includes declarative memory for properties of objects as well as episodic memory for events [44]. For example, the encoding of visual stimuli (photographs) was reported to be an expression of the memory-based looking behavior that evolves as early as at one year of age [45]. Moreover, the notions of self and of language have been shown to support the emergence of mature autobiographical memory [46]. This development has been ascribed to changes and growth in neural connectivity in the brain. Conversely, memories of events from early in infancy and early childhood can barely be recalled later in life (**Figure 2**). This has been termed infantile amnesia and ascribed to high rates of hippocampal neurogenesis [46]. This high neurogenesis was found to compromise the manifestation of original memory traces by causing an inability to form lasting memories [6, 45]. Conversely, there is evidence that "forgotten" early-acquired memory traces can affect a variety of behavioral responses later in life [47]. New empirical studies, however, suggest that people can recall a lot of memories from their preschool years, exceeding what had been published earlier [48]. Because memory recall has been described as independent of the framing context of the memories to be recalled [49], a recall of perceptual experiences is apparently independent from stating verbally what happened. Thus, the ability to recollect infantile memories apparently requires an age-dependent maturation of the brain [50].

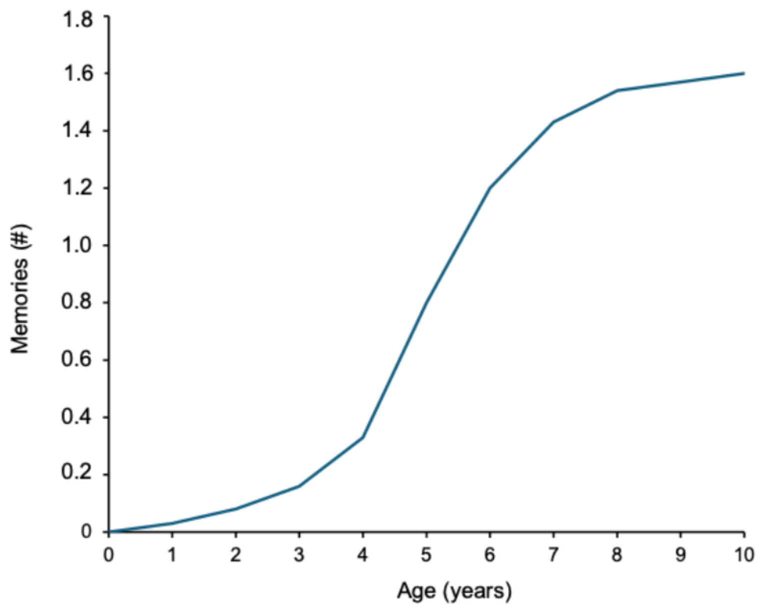

Figure 2. Interpolated graph showing the prominent increase in the number (#) of memories that can be stated in free recall after the age of four years (data from [6]).

4. Neural basis for the perceptuo-cognitive development

Following the description of the development of perceptual and cognitive processing in infancy and childhood the neural basis of this development will be presented in this section. There is evidence from in vivo magnetic resonance imaging that the human cortical microstructure involving myelin formation changes profoundly around birth [51]. Functional imaging has shown that information about social interactions is processed in the superior temporal gyrus in children as young as 3 years of age and evolves to an adult-like property by the age of 7 [52]. The online memory encoding of photographs during infancy has been related to greater activity in the hippocampus, as shown by functional magnetic resonance imaging [53]. Remembering sensory information such as familiar people was found in a small study to activate a widespread cortical network including the left posterior cingulate cortex, the left anterior orbitomedial cortex, the anterior middle prefrontal cortex, the precuneus, the posterior inferior parietal cortices, and the right posterior cingulate [53]. Moreover, recollection of externally experienced real events in autobiographical memory has been shown to involve the ventral medial prefrontal cortex as well as the retrosplenial cingulate cortex [54]. These networks do not include the parietal cortex, which is corroborated by the finding that patients with lesions of the parietal lobe exhibit a normal recollection of the context in which events were experienced [55]. Importantly, encoding of scenes that later caused flashbacks was associated with activation of the amygdala, striatum, thalamus, rostral anterior cingulate cortex, and ventral occipital cortex [56]. However, flashbacks in acute traumatic stress disorder involve the posterior cingulate gyrus [57]. It ought to be mentioned that cortical activity is differentially related to that in the subcortical structures. For example, resting-state imaging in adolescents and young adults revealed that enhanced functional connectivity of the ventral striatum with the subgenual cingulate and medial orbitofrontal cortex was related to reward processing and decreased between these areas related to executive functions [58]. The dorsal striatum that is preferentially related to sensorimotor processing showed the opposite pattern.

An important domain concerns autobiographic narratives that become encoded in memory. Recently, meta-analyses encompassing a large number of different imaging studies and employing point-by-point statistical likelihood estimations of the original imaging data have shown that the left hippocampus and the posterior cingulate cortex are central structures related to the construction and subsequent active elaboration of autobiographical memories [59]. In addition, there were differences, such as the fact that the ventromedial prefrontal cortex and the left angular gyrus, right hippocampus, and precuneus were found to participate in the construction, while the right inferior frontal gyrus was active in the elaboration of autobiographical memory. Furthermore, recollection of recent episodic autobiographical memories activated the angular gyrus, dorsomedial prefrontal cortex, and the hippocampus, whereas remote episodic autobiographical memories activated the posterior cingulate cortex [60]. This is supplemented by findings that showed that recall of happy life events engaged beyond the hippocampal formation also the anterior medial prefrontal gyrus and cingulate cortex as well as the basal ganglia and thalamus [61]. This contrasts to conceptual knowledge that was reported to be related to a bilateral activation of the anterior temporal lobes with a left lateralization for written words or word retrieval [62]. It should be emphasized that prefrontal cortical networks contribute only indirectly to memory functions as they carry out higher order algorithms that transform immediate sensory, memory, and internal information to perform flexible functions such as inference, planning, delayed responses or rule use [63].

5. Development of conceptual constructs in infancy

As described above, infants experience their mothers or caregivers with many different attributes. Most importantly, the babies experience the physical warmth of their mother’s body and the hunger-stilling milk their mothers give to them. They also are likely to notice the transient character of the availability of milk in the individual feeding episodes. Also, babies typically intensely watch their mothers and learn to discriminate the emotional face expressions of their mother. In addition, they notice that other people turn to them and approach them typically with friendly inclination. As infants encode this multifaceted information coming from their surroundings, they apparently begin to establish an integrated concept of "mother" that becomes manifest as early as at approximately two months of age and a concept of "self" by around six months (**Figure 1**). Also, young children have been reported to learn from others by evaluating their competence and reliability, as well as by interpreting communicative acts they observe [64]. In consequence, they apparently trust those who are truthful and well-meaning, and who keep their promises [28]. Notably, these processes seem to manifest in a prelinguistic manner and long before children become verbally able to articulate their memories [6].

In addition, infants hear communication with them as well as that between other people and, thus, learn that language transmits information. Furthermore, infants are probably accustomed to combining varied perceptual and emotional information into a comprehensive concept of a feeding, caring, and communicating subject that is labeled with one of the first words infants can utter themselves, namely mom, mama or papa. Mother or father is the first superordinate concept infants typically develop comprising empirical, relational and conceptual contents [65]. Notably, this representation becomes associated with higher emotions such as trust, safety, satiety, encouragement and awe [66]. To build such emotionally loaded supraordinate concepts requires the capacity to recall this composite information from long-term memory and hold it online—at least partially—in conscious awareness for an extended period of time [65].

In this manner, infants can be hypothesized to acquire the concept of "self". This notion involves awareness of bodily integrity, including interoceptive sensations related to bodily functions, such as production of saliva, urine, and stools, the outpouring of blood upon injury, and emotional states like hunger, pain, and discomfort and their temporal durations [67]. Importantly, all these experiences are heavily emotionally charged. Over time, infants also recognize that they possess individual wishes and intentions that may differ from those of others [1]. In this way, infants seem to learn that certain items belong to themselves and not to others and develop a notion of self that can be articulated and communicated to other people. In other words, they appear to develop a superordinate "me"-concept that combines physical properties, transient events, and narratives [65].

6. Relation of perceptuo-cognitive abilities to belief formation in infancy and childhood

In the previous section we described that the conceptual constructs infants develop may be strongly loaded with emotions. Thereby, conceptual constructs accord with the credition theory on the nature of beliefs and believing [68]. We would like to point out that beliefs have been defined to represent external information in a probabilistic and emotionally charged fashion as depicted in **Figure 3**. The encounters of an individual in the environment are encoded in memory [69]. Beliefs prompt individuals' actions and attitudes and thereby facilitate someone behaving consistently and in a predictable manner over time [70]. Beliefs become stabilized and, thereby, determine an individual's behavior by reinforcement learning that is known to occur through subcortical brain circuits involving the cerebellum and the basal ganglia [71]. Thus, from a conceptual perspective, beliefs link the past of an individual with that individual's future. Full epistemological status as a belief, however, requires that such constructs are associated with the articulation of a predictive proposition whose validity and truth content is presupposed by the subject [72]. In contrast, initial infantile beliefs do not appear to be bound to language but rather correspond to the primal beliefs that in adults manifest in a prelinguistic fashion concerning perceived properties of objects and events [73].

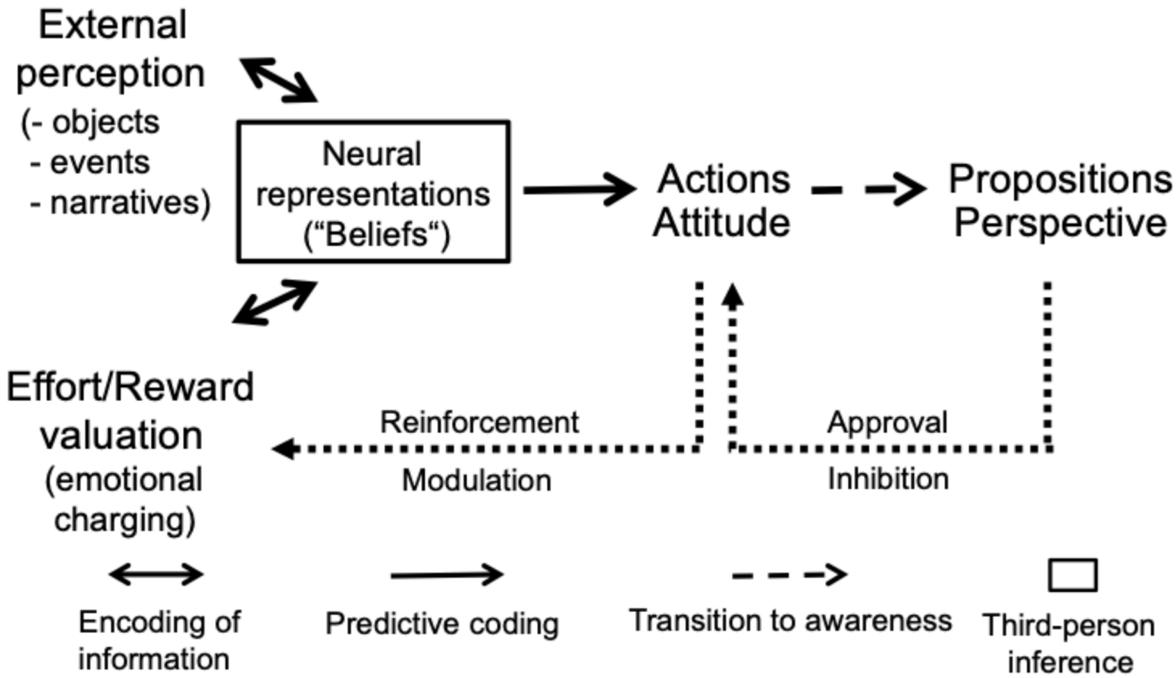

Figure 3. Model of belief formation and updating consisting of the neuro-cognitive processes that occur outside conscious awareness (**left side**) and of intentional verbalization and reflective processing of what one believes (**right side**). Belief formation results from the integration of perception of external information with the affective state of the individual affording the generation of actions and attitudes. Beliefs are enhanced by reinforcement learning or are modified in case of relevant prediction errors. After a subject becomes aware of what he/she believes, this can be expressed verbally as a proposition and determines the subject's perspectives. Dotted arrows signify feedback processing. Further developed from [68].

It is essential for beliefs that the multifaceted sensory information that characterizes any encounter concerning physical properties, episodic context, and associated causal explanations becomes affectively charged in terms of their subjective relevance (**Figure 3**). This emotional charging occurs fast and outside conscious awareness as it results from anxiety causing reflexive fleeing or, from the opposite, namely pleasure, which may lead to indulging in the given situation [74]. Accordingly, beliefs link perception to behavior such that the actions and attitude in response to threat or pleasure evolve within a fraction of a second. Moreover, beliefs may be modulated when new relevant information deviates from pre-existing belief-based predictions. The right-hand side of **Figure 3** shows that subjects can become aware of what they believe, develop a personal perspective, articulate it and communicate it to other people [75, 76]. This also allows the individual to reflect upon his/her own behavior and perspectives and to control the behavior voluntarily [65]. This accords with the notion that positive and negative emotions have been shown to make memories easier to retrieve, more richly experienced upon recollection, and more likely to influence behavior [77]. Moreover, the emotional context enhances the recollection of items from memory [7]. Notably, however, **Figure 3** signifies that a belief is a verbal term for the putative reason that someone infers from what another person has done or said [68].

Human behavior comprehends short-term and long-term goals that are characteristic for an individual. In fact, such an attitude has been described to be determined by both the properties of external stimuli and the subjective emotion they induce [78]. In other words, attitude is a personal expression of an underlying belief that capitalizes on the multitude of highly differentiated emotions including such as awe [79]. Like beliefs, attitudes manifest outside conscious awareness as depicted in **Figure 3**. When infants around six months of age [19] become aware of themselves, they begin to employ a first-person perspective (**Figure 3**). Only in their second year, however, do infants become aware of what they are doing as well as how and why [80, 81]. As soon as they can articulate this, they use verbs to describe their actions through utterances like “I do”, “I believe”, “I trust”, “I know”, etc., that have been argued to be universals [82]. Also, children develop an introspection concerning their own feelings (**Table 1**). Introspection into one’s momentary emotional state evolves further during childhood and is particularly influential in adolescence [83]. It can make people aware of their emotional state and describe the reason that caused it. Moreover, the personal engagement caused by a certain environmental condition is characterized by the person’s hope that something desirable may be achieved in the future [84]. Likewise, an individual may be in grief and mourn about a lost person or opportunity. Furthermore, empathy and theory of mind have been reported to occur in the age range from three to six years [19, 26]. This allows for a perspective towards other people (**Table 1**). Such a commitment towards other people and their affairs is the basis for understanding the emotional state of a companion and the reasons for their behavior [85]. When children pick up the meaning of narratives, they learn that people are expected to adhere to social norms and rules that are usually formalized in verbal scripts. By playing games, e.g., with their parents, siblings or others, children typically learn to comply to such norms and rules. Thereby children can be predicted to attain the notion of responsibility. Conversely, an individual’s aversion may cause an individual not to care about other persons’ affairs or even to hate them.

Infantile beliefs are hardly static and, similarly to beliefs in adults, likely to be modified in response to new information that contradicts or obviates prior experiences [69]. These processes can be expected to be particularly pronounced during infancy and childhood due to the extensive postnatal perceptive and cognitive development driven by the rapid growth and reorganization of the human brain. Thus, children’s experiences become increasingly differentiated and accurate. As described above, primal beliefs about objects and events are likely to manifest in infants and children in a similar fashion as in adults and allow for interpreting future encounters in a predictive manner [73]. Therefore, primal beliefs correspond to primals, priors and “bias” as described by Imhoff & Oeberst [86]. Notably, however, primal beliefs should not be confused with primal world beliefs [87]. In contrast to primal beliefs, the so-called primal world beliefs involve the semantic term “world” that has been shown to be a superordinate concept [65] and attributes such as “safe” or “dangerous”. These semantic attributes are dichotomous being inferences arrived at by conscious awareness.

Belief formation in children older than four years becomes increasingly complex, as children are exposed not only to perceptual experience but also more and more to narratives of different sources. Accordingly, social–emotional representations have been shown to be less differentiated in children aged six to twelve years than in adults [88]. In fact, it takes children a number of years to become competent in understanding social–emotional aspects of verbal statements. For example, children’s ability to evaluate testimony with regard to intent, knowledgeability, and trustworthiness increases over time [64]. Likewise, children of four years of age already have good theory of mind capabilities; they have been shown to be more proficient in reasoning about other persons’ intent or helpfulness than children at three years old [89]. Importantly, infants and children notice when other people share their beliefs and prefer individuals who like what they like [5].

Table 1. Subjective perspectives and corresponding verbal expressions as developed in childhood.

Oneself	**Verbal descriptions**
Introspection:	I am delighted to…
	I am interested in…
	I am prepared to…
	I am convinced that…
	I am confident that…
	I am afraid of…
	I am upset by…
	I am disgusted by…
	I am desperate because…
Personal engagement:	
Hope	I hope that…
Grief	I mourn because…
Commitment to other people and their affairs:	
Empathy	I share your joy that…
	I feel sorry for you that…
Theory of mind	I understand that you…
Responsibility	I accept our norms and rules that…
Aversion:	
Neglect	I do not care that…
Antipathy	I hate you for…

This list of verbal descriptions is not exhaustive.

7. Discussion

During the first four years of life, humans undergo a profound developmental trajectory concerning the acquisition of perceptual information and cognition that is fundamental for determining individuals' behavior. Humans rely on this over-arching framework not only in childhood but also during their entire life—in particular, also during adulthood. The neural circuits in the brain related to sensory processing develop during fetal life and continue to mature after birth [90]. These changes evolve in a use-dependent manner that has been attributed to ontogenetic neuroplasticity—a process that is particularly prominent during infancy and childhood but continues throughout adult life [20]. Under typical conditions, the human sensory systems provide reliable information about external states. Moreover, infantile exploratory object manipulation has been reported to stimulate cognitive development and predicts later cognitive performance [91]. Thus, sensory perceptions become more differentiated and developed in relation to the evolution of an infant's exploratory skills. In addition, infants appear to learn in parallel that their perceptions may be emotionally charged and have relevance to them. Emotional categories for information such as appealing, frightening, annoying, and disgusting can be deciphered reliably by infants at the age of 10 to 12 months [92]. Importantly, infants and children seemingly believe their emotionally charged perceptions and trust them intuitively as is most evident from their attitude towards their caregivers [1]. As infants lack linguistic abilities, these early beliefs can be expected to evolve in a prelinguistic manner but may be expressed verbally once sufficient language capacities have developed. Although most information acquired during the first four years of life cannot be recalled later during adolescence and adulthood, emotionally salient experiences are more likely to be retrieved from memory and be expressed in the format of propositions [31]. Such early beliefs may correspond to what has been called a priori knowledge, as coined by philosophy, and determine an individual's behavior and perspectives on the world.

The acquisition of information concerns various aspects such as the multimodal dimensions of objects and dedicated interactions of infants with objects, other humans and animals, as well as exposure to narratives. The more frequently infants and children are exposed to these different kinds of information, the more confident they become that they are true and trust and believe them [1]. It can be hypothesized that children, like adults, have a high competence for developing trust when the information is accompanied by a prominent emotional engagement. Only much later, however, when children reach the age of eight to thirteen years, watching of video clips can evoke the emotion of awe as indicated by an exhibited increased respiratory sinus arrhythmia that is an index of parasympathetic nervous system activation associated with emotional engagement [93]. Furthermore, it is well known that only in puberty and adolescence do the emotional, psychosocial and cognitive capacities develop further [83]. This development has been reported to include a number of capabilities such as the emergence of abstract thinking, a growing ability to adopt the perspectives and viewpoints of others, an increased ability of introspection, the establishment of values, and the emergence of coping strategies to overcome problems [94]. It should be acknowledged, though, that fluid intelligence, namely fast and logical problem solving, peaks at the age of about 20 years, while the ability to make use of acquired knowledge and skills (crystalized intelligence) continues to increase in capacity up to the age of 40 to 65 years, with particular emphasis on emotional intelligence and core personality traits [95]. Accordingly, beliefs can be hypothesized to change and thus become progressively differentiated during a person's lifetime.

Similarly to adults, primal beliefs about objects and events acquired in infancy and childhood are formed rapidly, signal an individual's experience and possess a truth value only for the individual who holds them [68]. In contrast, conceptual beliefs, including autobiographical, religious, political and science-based beliefs, are grounded in narratives that require longer episodes for apprehension than, for example, tactile or visual exploration of objects. Therefore, comprehending the content of a narrative depends heavily on the working memory capacity and the conceptual properties of infants and children. Notably, children appear to conceptualize unobservable scientific information and religious entities in similar ways [96]. Likewise, for children aged three to eight, Santa Claus and God share some concrete attributes but seem to differ in those that are more abstract [97]. Accordingly, children seem to be imprinted with a God image from an early age onwards and may retain the belief into adulthood. While these processes appear to be universal, their contents may vary across different socio-cultural environments and are likely to be related to different rituals. For example, a between-ethno-religious-group comparison of Hindu people in Mauritius found evidence supporting the notion that rituals play an important role in human societies as mechanisms for managing perceptions of costs and enhancing social cohesion and emotional well-being [98]. Such behavioral manifestations definitely go beyond the sensorimotor system and encompass other systems such as affect, mentalizing and tracking analogs that can be differentiated linguistically [99] but are integral components of beliefs. Upbringing in a religious environment, compared to upbringing in a non-religious environment, however, does not affect executive functions in adulthood such as cognitive flexibility that are known to be mediated by the prefrontal cortex. In contrast, active religious practice in adulthood has been associated with reduced cognitive flexibility as measured with standard neuropsychological tests [100]. Accordingly, the experience of children varies not only between their families but also today with respect to what they are exposed to via digital equipment. One may hypothesize that this may explain the growing heterogeneity of religious and political beliefs in contemporary Western societies.

8. Conclusions

This work demonstrates that early in infancy prelinguistic, emotionally charged experiences of objects and events induce the first primal beliefs. As language capacities develop, children become able to recall their perceptual encounters and to verbally express the contents of their thoughts and their subjective perspectives towards themselves and other people, including their affairs. Understanding these processes is crucial, as this provides the foundational scaffolding for later cognitive, social, and cultural belief systems that are likely to shape human behavior across the lifespan. Longitudinal studies may shed further light on the role of these processes.

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Conflict of interest

The authors declare that they have no competing interests.

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